

The Arrhenius Model 1

Essential Pre-Uni Chemistry M3.1

Subject & topics: Chemistry | Physical | Kinetics Stage & difficulty: A Level P2

$$k = Ae^{-E_A/RT}$$

An Arrhenius plot is a graph of $\ln(k)$ against $\frac{1}{T}$ in K^{-1} .

Part A

y-intercept

On a plot of $\ln(k)$ against $\frac{1}{T}$, what is the y -intercept?

The following symbols may be useful: A, E_A , R, T, k

Part B

Units of gradient

Give the units of the gradient of an Arrhenius plot.

- ☐ K m^{-2}
- ☐ K
- ☐ $^{\circ}\text{C}$
- ☐ N dm^{-3}

Part C

Gradient

On a plot of $\ln(k)$ against $\frac{1}{T}$, what is the gradient?

The following symbols may be useful: A, E_A , R, T, k

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The Arrhenius Model 2

Essential Pre-Uni Chemistry M3.2

Subject & topics: Chemistry | Physical | Kinetics Stage & difficulty: A Level P2

$$k = Ae^{-E_A/RT}$$

An Arrhenius plot is a graph of $\ln(k)$ against $\frac{1}{T}$ in K^{-1} .

Part A

Activation energy

If the gradient of an Arrhenius plot is -1203 K , find the activation energy. Use $R = 8.3145 \text{ J mol}^{-1} \text{ K}^{-1}$.

Part B

Activation energy II

If the gradient of an Arrhenius plot is -4250 K , find the activation energy. Give your answer to 3 significant figures.

Part C

Gradient of Arrhenius plot

If a reaction has activation energy of 16.5 kJ mol^{-1} , find the expected gradient of an Arrhenius plot.

Part D

***y*-intercept**

The pre-exponential factor, A , is found to have a value of 0.6 s^{-1} for a first-order reaction. Calculate the expected *y*-intercept of an Arrhenius plot.

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The Arrhenius Model 3

Essential Pre-Uni Chemistry M3.3

Subject & topics: Chemistry | Physical | Kinetics **Stage & difficulty:** A Level P2

$$k = Ae^{-E_A/RT}$$

An Arrhenius plot is a graph of $\ln(k)$ against $\frac{1}{T}$ in K^{-1} .

Part A

A for a first-order reaction

The y -intercept of an Arrhenius plot for a first-order reaction is at -2.30 . Find the pre-exponential factor, A , according to the Arrhenius model.

Part B

A for a second-order reaction

The y -intercept of an Arrhenius plot for a second-order reaction is at 3.20 . Find the pre-exponential factor, A , according to the Arrhenius model.

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The Arrhenius Model 4

Essential Pre-Uni Chemistry M3.4

Subject & topics: Chemistry | Physical | Kinetics Stage & difficulty: A Level P3

$$k = Ae^{-E_A/RT}$$

An Arrhenius plot is a graph of $\ln(k)$ against $\frac{1}{T}$ in K^{-1} .

Part A

Activation energy

The rate constant, k , for a first-order reaction is found to be 0.0250 s^{-1} at 290 K . If the pre-exponential factor is 26.0 s^{-1} , find the activation energy.

Part B

Pre-exponential factor A

The rate constant, k , for a second-order reaction is found to be $0.050 \text{ dm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ at 300 K . If the activation energy is 2.50 kJ mol^{-1} , find the value of the pre-exponential factor, A .

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The Arrhenius Model 6

Essential Pre-Uni Chemistry M3.6

Subject & topics: Chemistry | Physical | Kinetics **Stage & difficulty:** A Level P3

$$k = Ae^{-E_A/RT}$$

An Arrhenius plot is a graph of $\ln(k)$ against $\frac{1}{T}$ in K^{-1} .

If a reaction has activation energy 14.0 kJ mol^{-1} , and a pre-exponential factor of 120 s^{-1} , find the temperature at which the rate constant is equal to 2.00 s^{-1} .

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The Arrhenius Model 7

Essential Pre-Uni Chemistry M3.7

Subject & topics: Chemistry | Physical | Kinetics Stage & difficulty: A Level P3

$$k = Ae^{-E_A/RT}$$

An Arrhenius plot is a graph of $\ln(k)$ against $\frac{1}{T}$ in K^{-1} .

A reaction is found to have a rate constant of $1.25 \times 10^{-3} \text{ dm}^6 \text{ mol}^{-2} \text{ s}^{-1}$ at 400 K and $1.60 \times 10^{-3} \text{ dm}^6 \text{ mol}^{-2} \text{ s}^{-1}$ at 500 K.

Part A

E_A

Find the activation energy.

Part B

A

Find the pre-exponential factor, A .

Part C

Order of the reaction

Give the overall order of reaction.

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The Arrhenius Model 8

Essential Pre-Uni Chemistry M3.8

Subject & topics: Chemistry | Physical | Kinetics Stage & difficulty: A Level P2

$$k = Ae^{-E_A/RT}$$

An Arrhenius plot is a graph of $\ln(k)$ against $\frac{1}{T}$ in K^{-1} .

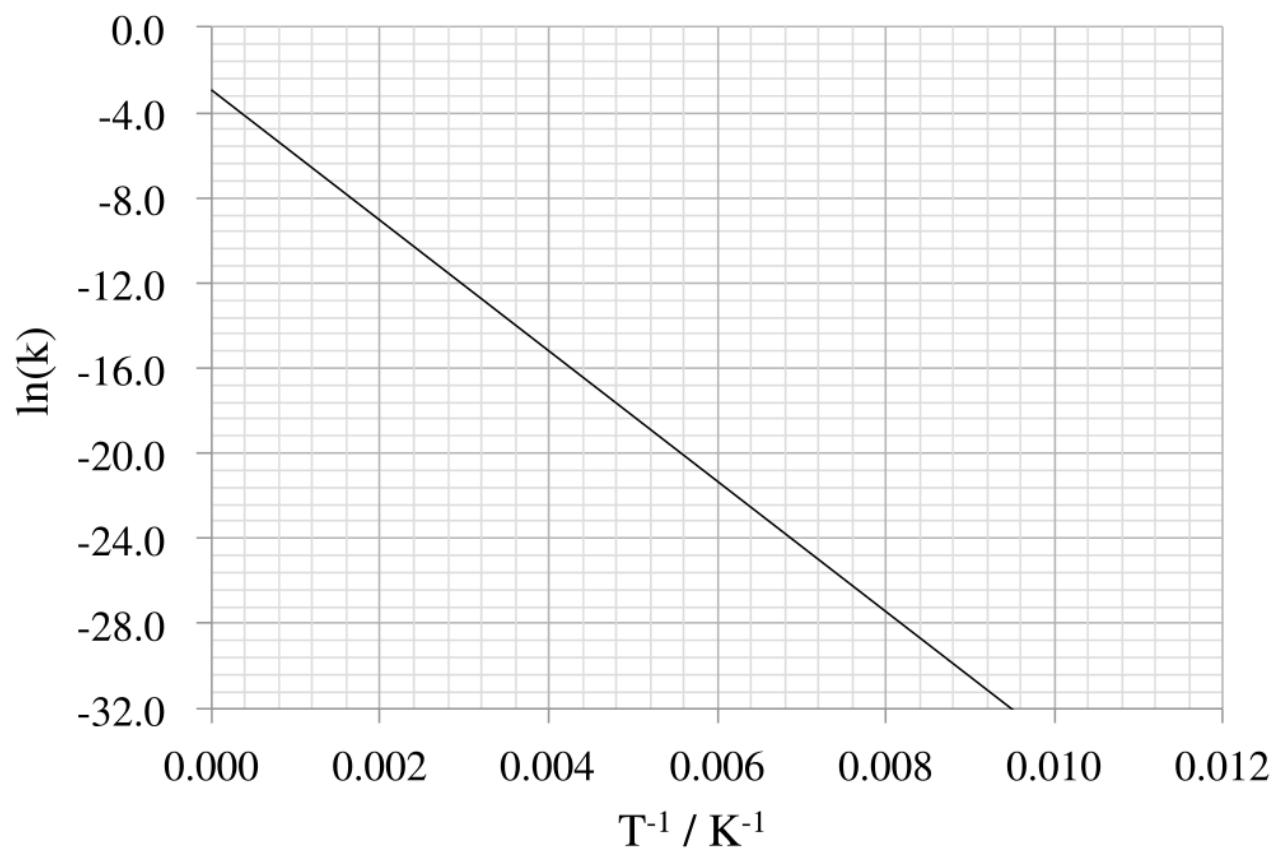


Figure 1: M3.8 Arrhenius Plot

Part A

E_A

Using the graph above, find the activation energy. Give your answer to 2 significant figures

Part B

A

Using the graph above, find the pre-exponential factor. Give your answer to 1 significant figure

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Catalysis 1

Essential Pre-Uni Chemistry M4.1

Subject & topics: Chemistry | Physical | Kinetics **Stage & difficulty:** A Level P1

The iodination of propanone, $\text{C}_3\text{H}_6\text{O} + \text{I}_2 \longrightarrow \text{C}_3\text{H}_5\text{OI} + \text{HI}$, when catalysed in aqueous conditions, obeys the rate law:

$$\text{rate} = k[\text{C}_3\text{H}_6\text{O}][\text{HCl}]$$

Part A

Catalyst

Identify the catalyst in this reaction.

Part B

Type of catalyst

Is the catalyst homogeneous or heterogeneous?

- ☐ homogeneous
- ☐ heterogeneous

Part C

Concentration of catalyst

If the catalyst has an initial concentration of $0.020 \text{ mol dm}^{-3}$, give the concentration of the catalyst when the concentration of propanone has decreased to one quarter of its original value.

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Catalysis 6

Essential Pre-Uni Chemistry M4.6

Subject & topics: Chemistry | Physical | Kinetics **Stage & difficulty:** A Level P1

Complete the following description of heterogeneous catalysis.

A heterogeneous catalyst works best when it has a large so that many of its are exposed to the reactants. In the first stage, reactants are onto the catalytic surface, weakening internal bonds in the reactant particles. In the second stage, the particles react to form products. The of the slowest step in this reaction is lower than that of the slowest step in the uncatalysed reaction. The third stage is the , or release, of product particles from the surface. This the surface ready for further reactions. If the products are not released, or if some contaminant binds to the surface, further catalytic activity is impeded and the catalyst has been .

Items:

mass desorption enthalpy change rate density destruction poisoned absorbed regenerates
 surface area active sites volume adsorbed activation energy corroded

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Arrhenius Equation

Subject & topics: Chemistry | Physical | Kinetics **Stage & difficulty:** A Level C2

The temperature of a sample of hydrogen iodide is raised from 300°C to 500°C . The activation energy for the dissociation of hydrogen iodide ($2\text{HI} \rightarrow \text{H}_2 + \text{I}_2$) is 190 kJ mol^{-1} .

Part A

Rate constants

Calculate the ratio of the rate constants at the two temperatures (as a number greater than 1, to 2 significant figures).

Part B

Average velocity

Calculate the ratio of the average velocities of the molecules at the two temperatures (to 2 significant figures).

Part C

Activation energy

The enzyme glucose-6-phosphate dehydrogenase catalyses the reaction:

Given the data below and the Arrhenius equation $k = Ae^{-\frac{E_a}{RT}}$, use a graphical method to calculate the activation energy of the reaction, E_a .

$T / ^\circ\text{C}$	$k / \text{M}^{-1} \text{s}^{-1}$
30	1.95
35	2.40
40	2.82
45	3.31
50	3.89

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